

Population Differentiation Lab

User manual for teaching population ecology and genetics

This manual describes how the web application works, how to enter parameters, how to interpret outputs, and which assumptions, equations, and random distributions are used by the model.

1. Purpose

Population Differentiation Lab is a browser-based teaching tool for exploring how local population size, mutation, selection, migration, extinction, drift, and local mating assumptions affect genetic differentiation in a metapopulation.

The app is designed for conceptual exploration. It is not intended to replace a full individual-based population-genetic simulator or a formal inference package.

2. Running The Program

1. Open `index.html` in a web browser.
2. Enter model parameters in the input panels.
3. Press **Run model**.
4. Inspect summary values, tables, dynamics graphs, and fixation outcomes.
5. Use **Reset** to return to default settings, or **Load example** to load a demonstration metapopulation.
6. Use **Neutral drift check** to load an isolated neutral two-population diagnostic scenario.

3. Inputs

Input	Meaning
Local populations	Number of local populations. The default is 2; more can be added.
Number of loci	Total loci represented in heterozygosity calculations. Allele dynamics are shown for five tracked loci.
Isolation time, t	Number of generations used for dynamic calculations.
Mutation rate	Symmetric mutation rate between the tracked allele and the alternative allele.

Input	Meaning
Permutations	Number of stochastic replicate simulations.
Local mating model	Controls how Fis is generated: Hardy-Weinberg, partial inbreeding, or cumulative drift.
Inbreeding rate	Used only in the partial inbreeding model.
Environmental synchrony	Correlation of environmental noise among populations. 0 means independent local shocks; 1 means fully shared environmental shocks.
Selection coefficients	For the five tracked loci. Positive s lowers tracked-allele fitness as $w = 1 - s$.
Initial allele frequencies	Starting frequency of one tracked allele at five loci in each local population.
N0	Initial population size at generation 0.
Growth rate r	Intrinsic growth rate in the stochastic logistic population-size model.
Environmental noise SD	Standard deviation of environmental noise added to the logistic growth exponent.
K	Upper limit for local population size.
Migration matrix	Directional migration rates. Rows are source populations and columns are destination populations.
Stepping-stone indirect migration for Fst	When checked, indirect migration paths are included in the migration modifier for pair-wise Fst.
Founder-sample initial allele copies	When checked, initial allele frequencies are converted to independent binomial allele-copy samples for each population and locus.
Migration changes population size	When checked, realized migrants also move individuals among populations and can rescue or recolonize demes.

4. Population-Size Model

Each permutation starts with local population size . For each later generation, population size follows a stochastic logistic update, rounded to an integer, and bounded between 0 and K.

```

shared shock ~ Normal(0, 1)
local shock ~ Normal(0, 1)
environmental noise = SD * (sqrt(c) * shared shock + sqrt(1 - c) * local
shock)
N(t+1) = N(t) * exp( r * (1 - N(t) / K) + environmental noise )
N(t+1) = round( clamp( N(t+1), 0, K ) )

```

If $N_t < 2$, that local population is considered extinct for the rest of that permutation unless demographic migration later recolonizes it.

- Extinct populations have allele frequency 0 in the plots.
- Extinct populations have $H_{exp} = 0$ and $H_{obs} = 0$.
- Extinct populations do not send migrants.
- Extinct populations are excluded from comparable pair-wise F_{st} dynamics.
- If demographic migration is enabled, incoming migrants can rescue or recolonize a deme once local N reaches at least 2.

5. Migration Model

Migration rates are directional. For example, the rate from population 1 to population 2 can differ from the rate from population 2 to population 1.

Realized migrants are integer counts drawn from a Poisson distribution:

```

expected migrants from i to j =  $N_i * m_{ij}$ 
 $M_{ij} \sim \text{Poisson}(N_i * m_{ij})$ 

```

This permits rare migration. For example, if $N = 100$ and $m = 0.001$, then expected migrants per generation are 0.1, or about one migrant every ten generations on average.

Stepping-Stone Indirect Migration For F_{st}

If the stepping-stone option is enabled, pair-wise F_{st} can include indirect migration paths. Indirect path contributions are calculated as products of the direct rates along a path.

```

 $m_{13, \text{indirect}} = m_{12} * m_{23}$ 

```

Thus, if $m_{12} = 0.1$ and $m_{23} = 0.1$, then the indirect contribution from population 1 to population 3 is 0.01.

Allele movement is still simulated generation by generation using direct migration. Indirect gene flow appears through intermediate populations over time. The indirect-path option affects the F_{st}

migration modifier.

6. Allele Dynamics

The app tracks one allele at five loci in each population. Let p be the frequency of the tracked allele.

Selection

The tracked allele has relative fitness $w = 1 - s$. The alternative allele has fitness 1. Positive s therefore selects against the tracked allele.

```
w_tracked = max(0, 1 - s)
w_alternative = 1
mean_fitness = p * w_tracked + (1 - p) * w_alternative
p_after_selection = (p * w_tracked) / mean_fitness
```

Migration

After selection, destination populations receive realized integer migrants. The allele frequency after migration is a weighted average of resident and immigrant allele frequencies.

```
p_after_migration =
((resident count * p_resident) + sum(migrants from source i * p_i)) /
N_destination
```

Mutation

Mutation is symmetric between the tracked allele and alternative allele.

```
p_after_mutation = p * (1 - mutation_rate) + (1 - p) * mutation_rate
```

Genetic Drift

If founder sampling is enabled, the initial allele frequency entered by the user is treated as an expected frequency. Initial allele copies are sampled independently for each population and locus.

```
X0 ~ Binomial(2N0, p0)
p0_realized = X0 / (2N0)
```

Genetic drift is modeled by Wright-Fisher binomial sampling of gene copies.

```
X ~ Binomial(2N, p_after_mutation)
p_next = X / (2N)
```

Because this final step is discrete, tracked alleles can become fixed at frequency 1 or extinct at frequency 0.

7. F-Statistics

Fis

Fis is treated as a within-population quantity. It measures departure of observed heterozygosity from Hardy-Weinberg expectation.

$$F_{is} = 1 - H_{obs} / H_{exp}$$

The app offers three local Fis models:

Model	Equation	Interpretation
Hardy-Weinberg	$F_{is} = 0$	Random mating within each local population each generation.
Partial inbreeding	$F_{is}(t+1) = s * (1 + F_{is}(t)) / 2$	Fis approaches equilibrium $s / (2 - s)$.
Cumulative drift	$F_{is} = 1 - (1 - 1 / (2N_e))^t$	Closed finite population contrast where identity by descent accumulates.

Fst

Fst is not population-specific. It is calculated for population pairs, and summary Fst is the mean across pair-wise values.

$$\begin{aligned} F_{st_isolation} &= 1 - (1 - 1 / (2N_e))^t \\ F_{st_migration} &= 1 / (1 + 4Nm) \\ \text{Dynamic pair } F_{st} &= F_{st_isolation} * F_{st_migration} \end{aligned}$$

The migration expression is treated as an equilibrium modifier, not as a generation-0 value. Therefore pair-wise Fst starts at 0 when initial allele frequencies are equal.

Fit

Fit is calculated from Fis and Fst:

$$1 - F_{it} = (1 - F_{is}) * (1 - F_{st})$$

8. Heterozygosity

The app reports expected and observed heterozygosity for each local population. Hexp is derived from a mutation-drift teaching approximation using theta:

```
theta = 4 * Ne * mutation rate
neutral Hexp = theta / (1 + theta)
```

Selection and the number of loci apply additional teaching-model adjustments to Hexp. Hobs is then calculated from Fis:

```
Hobs = Hexp * (1 - Fis)
```

Low Hexp means low genetic diversity. High Fis means heterozygote deficit relative to Hardy-Weinberg expectations. These are not the same quantity.

9. Random Distributions Used

Process	Distribution	Use
Population size	Stochastic logistic update with shared/local environmental noise	Generates local N through time from the previous generation.
Migrant counts	Poisson(N_source * migration rate)	Generates realized integer migrants.
Genetic drift	Binomial(2N, allele frequency)	Wright-Fisher sampling of gene copies.
HPD intervals	Empirical shortest 95% interval from permutation values	Grey uncertainty bands in dynamics plots.

10. Outputs

Output	Meaning
Metapopulation summary	Mean Fis, mean pair-wise Fst, Fit, and mean Hexp/Hobs.
Local population outputs	Mean final Ne, Hexp, Hobs, Fis, 95% interval for Hexp, and extinction risk.

Output	Meaning
Pair-wise Fst table	Pair-specific isolation Fst, migration Fst, combined Fst, and Nm.
F-statistic dynamics	Fis per population and Fst per population pair, with 95% HPD bands.
Allele dynamics	Mean allele-frequency trajectories and 95% HPD bands for five tracked loci.
Fixation outcomes	Permutation proportions ending as same tracked fixed, same alternative fixed, total same fixed, different fixed, polymorphic, or extinct.

11. Interpreting Mean Allele-Frequency Plots

Mean allele-frequency lines can remain close to 0.5 even when individual permutations have fixed at 0 or 1. For this reason, the app also reports 95% HPD bands and fixation outcome proportions.

The fixation outcome table is often the clearest way to see whether two populations tend to fix the same allele, fix different alleles, remain polymorphic, or go extinct.

The Neutral drift check preset loads two isolated populations with equal initial allele frequencies, no mutation, no migration, no selection, constant small population size, and many generations. Under this independent neutral scenario, fixed outcomes should approach approximately:

```

same tracked fixed      = 0.25
same alternative fixed   = 0.25
different fixed         = 0.50

```

The "different among fixed" column is useful when some permutations are still polymorphic at the final generation.

12. Recommended Teaching Scenarios

- **Random mating, stable demes:** choose Hardy-Weinberg Fis, moderate-to-high N_0 , positive r , low environmental noise, and moderate-to-high K .
- **Drift and fixation:** use small N_e , low migration, low mutation, and many generations.
- **Rare migration:** use small migration rates such as 0.001; Poisson migration can produce occasional migrants.
- **Stepping-stone structure:** use three or more populations with migration between neighboring demes and enable indirect migration for Fst.
- **Inbreeding equilibrium:** choose partial inbreeding and vary the inbreeding rate.